

USER INFORMATION
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INFUSOL® PCM

Paracetamol 10mg/mL Intravenous Infusion

Sterile and Non-Pyrogenic

Composition:
Each mL solution for infusion contains 10mg Paracetamol BP.
Each 100mL bottle contains 1000mg Paracetamol BP.

Product Description:
INFUSOL® PCM is a slightly pinkish-orangish and practically free from particle.

Pharmacodynamic:
Mechanism of action
The precise mechanism of the analgesic and antipyretic properties of paracetamol has still to be established; it may involve central and peripheral actions.
Pharmacodynamic effects
Paracetamol provides onset of pain relief within 5 to 10 minutes after the start of administration. The peak analgesic effect is obtained in 1 hour and the duration of this effect is usually 4 to 6 hours. INFUSOL® PCM reduces fever within 30 minutes after the start of administration with a duration of the antipyretic effect of at least 6 hours.

Pharmacokinetics:
Adults
Absorption:
Paracetamol pharmacokinetics is linear up to 2g after single administration and after repeated administration during 24 hours. The bioavailability of paracetamol following infusion of 1g of INFUSOL® PCM is similar to that observed following infusion of 2g propacetamol (containing 1g paracetamol respectively). The maximal plasma concentration (C_{max}) of paracetamol observed at the end of 15 minutes intravenous infusion of 1g of INFUSOL® PCM is about 30µg/mL respectively.

Distribution:
The volume of distribution of paracetamol is approximately 1L/kg. Paracetamol is not extensively bound to plasma proteins. Following infusion of 1g paracetamol, significant concentrations of paracetamol (about 1.5µg/mL) were observed in the cerebrospinal fluid at and after the 20th minute following infusion.

Biotransformation:
Paracetamol is metabolised mainly in the liver following two major hepatic pathways: glucuronic acid conjugation and sulphuric acid conjugation. The latter route is rapidly saturable at doses that exceed the therapeutic doses. A small fraction (less than 4 %) is metabolised by cytochrome P450 to a reactive intermediate (N-acetyl benzoquinone imine) which, under normal conditions of use, is rapidly detoxified by reduced glutathione and eliminated in the urine after conjugation with cysteine and mercapturic acid. However, during massive overdosing, the quantity of this toxic metabolite is increased.

Elimination:
The metabolites of paracetamol are mainly excreted in the urine. 90 % of the dose administered is excreted within 24 hours, mainly as glucuronide (60 – 80 %) and sulphate (20 – 30 %) conjugates. Less than 5 % is eliminated unchanged. Plasma half-life is 2.7 hours and total body clearance is 18L/h.

Neonates, infants and children
The pharmacokinetic parameters of paracetamol observed in infants and children are similar to those observed in adults, except for the plasma half-life that is slightly shorter (1.5 to 2 h) than in adults. In neonates, the plasma half-life is longer than in infants i.e. around 3.5 hours. Neonates, infants and children up to 10 years excrete significantly less glucuronide and more sulfate conjugates than adults.

Table - Age related pharmacokinetic values (standardised clearance, *CL_{std}/F_{oral} (L.h⁻¹ 70kg⁻¹)

Age	Weight (kg)	CL _{std} /F _{oral} (L.h ⁻¹ 70kg ⁻¹)
40 weeks PCA	3.3	5.9
3 months PNA	6	8.8
6 months PNA	7.5	11.1
1 year PNA	10	13.6
2 years PNA	12	15.6
5 years PNA	20	16.3
8 years PNA	25	16.3

*CL_{std} is the population estimate for CL

Special populations
Renal insufficiency
In cases of severe renal impairment (creatinine clearance 10-30 mL/min), the elimination of paracetamol is slightly delayed, the elimination half-life ranging from 2 to 5.3 hours. For the glucuronide and sulfate conjugates, the elimination rate is 3 times slower in subjects with severe renal impairment than in healthy subjects. Therefore, when giving paracetamol to patients with severe renal impairment (creatinine clearance ≤30 mL/min), the minimum interval between each administration should be increased to 6 hours.

Elderly subjects
The pharmacokinetics and the metabolism of paracetamol are not modified in elderly subjects. No dose adjustment is required in this population.

Indication:
Paracetamol 10mg/mL Intravenous Infusion is indicated for the short-term treatment of moderate pain, especially following surgery, and for the short-term treatment of fever, when administration by intravenous route is clinically justified by an urgent need to treat pain or hyperthermia and/or when other routes of administration are not possible.

Recommended Dosage:
The 100 ml bottle is restricted to adults, adolescents and children weighing more than 33 kg.

Posology:
The dose to be administered and the container size to be used depend exclusively on the patient’s weight. The volume to be administered must not exceed the determined dose. If applicable the desired volume must be diluted in a suitable solution for infusion prior to administration or a syringe driver must be used. Dosing based on patient weight (please see the dosing table here below):

Patient weight (kg)	Dose/administration	Volume/administration	Maximum volume of INFUSOL® PCM per administration based on upper weight limits of group (mL)***	Maximum daily dose**
≤ 10kg*	7.5mg/kg	0.75mL/kg	7.5mL	30mg/kg
> 10kg to ≤ 33kg	15mg/kg	1.5mL/kg	49.5mL	60mg/kg < 2g
> 33kg to ≤ 50kg	15mg/kg	1.5mL/kg	75mL	60mg/kg < 3g
> 50kg with additional risk factors for hepatotoxicity	1g	100mL	100mL	3g
> 50kg and no additional risk factors for hepatotoxicity	1g	100mL	100mL	3g

** Preterm newborn infants:
No safety and efficacy data are available for premature newborn infants*
*** Maximum daily dose:
The maximum daily dose as presented in the table above is for patients that are not receiving other paracetamol containing products and should be adjusted accordingly taking such products into account.*
**** Patients weighing less will require smaller volumes.
The minimum interval between each administration must be at least 4 hours. The minimum interval between each administration in patients with severe renal insufficiency must be at least 6 hours.
No more than 4 doses to be given in 24 hours.*

Severe renal insufficiency:
It is recommended, when giving paracetamol to patients with severe renal impairment (creatinine clearance ≤ 30mL/min), to increase the minimum interval between each administration to 6 hours.
Adults with hepatocellular insufficiency, chronic alcoholism, chronic malnutrition (low reserves of hepatic glutathione), dehydration: The maximum daily dose must not exceed 3000mg.

Method of administration
Take care when prescribing and administering INFUSOL® PCM to avoid dosing errors due to confusion between milligram (mg) and millilitre (mL), which could result in accidental overdose and death. Take care to ensure the proper dose is communicated and dispensed. When writing prescriptions, include both the total dose in mg and the total dose in volume. Take care to ensure the dose is measured and administered accurately.
Intravenous use.
The paracetamol solution is administered as a 15-minute intravenous infusion.

Patients weighing ≤10kg:
• The volume to be administered should be withdrawn from the container and diluted in a sodium chloride 9mg/mL (0.9 %) solution or dextrose 50mg/mL (5 %) solution or a combination of both solutions up to one tenth (one volume INFUSOL® PCM into nine volumes diluent) and administered over 15 minutes.
• A 5 or 10mL syringe should be used to measure the dose as appropriate for the weight of the child and the desired volume. However, this should never exceed 7.5mL per dose.
• The user should be referred to the product information for dosing guidelines.

For single use only. Any unused solution should be discarded.
Before administration, the product should be visually inspected for any particulate matter and discolouration. Only to be used if solution is clear, colourless to slightly pinkish-orangish (perception may vary) and the container and its closure are undamaged. Colouration can intensify over time without adversely affecting the quality of the product.
As for all solutions for infusion presented in containers with air space inside, it should be remembered that close monitoring is needed notably at the end of the infusion, regardless of administration route. This monitoring at the end of the infusion applies particularly for central route infusions, in order to avoid air embolism.

Route of administration: Intravenous (I.V)

Contraindications:
Hypersensitivity to paracetamol and propacetamol HCl (prodrug of paracetamol). Severe hepatocellular insufficiency.

Warnings and Precautions:

WARNING
This preparation contains PARACETAMOL.
Do not take any other paracetamol containing medicines at the same time.

Allergy alert: Paracetamol may cause severe skin reactions. Symptoms may include skin reddening, blisters or rash. These could be signs of a serious condition. If these reactions occur, stop use and seek medical assistance right away.

